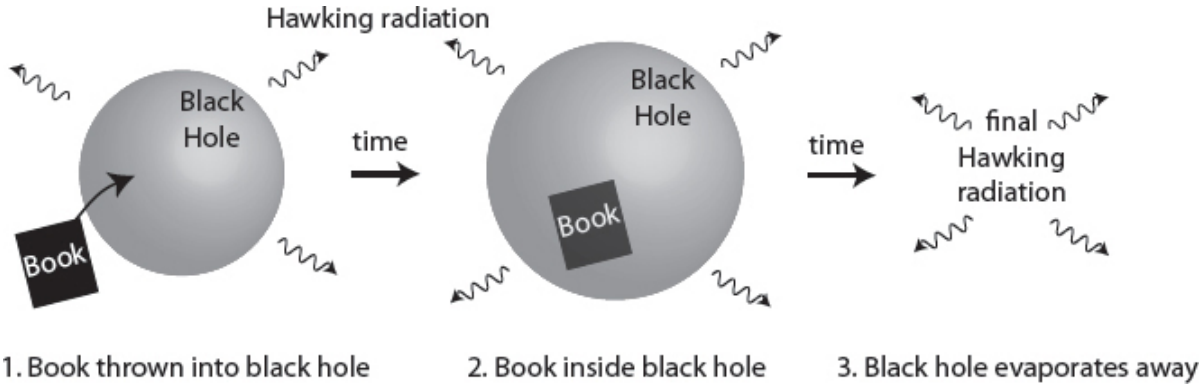
like an ordinary fire. It's not like we have any experimental input one way or the other. But most physicists believe that information is conserved, and that it really does get out somehow. And they suspect that the secret to getting it out lies in a better understanding of quantum gravity.

That's easier said than done. One way of thinking about why black holes are supposed to be black in the first place is that in order to escape, you would have to be able to travel faster than light. Hawking radiation avoids that difficulty because it actually originates right outside the event horizon, not deep in the interior. But any book we throw inside does indeed plunge into the interior, with all its information intact. You might wonder whether the information is somehow copied onto the outgoing radiation as the book falls through the horizon, and carried out that way. Unfortunately that's in contradiction with the basic principles of quantum mechanics; there is a result called the *no-cloning theorem* that says we can't duplicate quantum information without destroying the original copy.



The other possibility seems to be that the book falls all the way in, but as it hits the singularity inside the black hole, its information is somehow transferred to the outgoing radiation at the horizon. Unfortunately, that would seemingly require faster-than-light communication. Or, equivalently, dynamical nonlocality—occurrences at one point in spacetime immediately influencing what happens some distance away. This kind of nonlocality is precisely what cannot happen, according to the ordinary rules of quantum field theory. This is a clue that those rules might have to be dramatically revised once quantum gravity becomes important.*